Quantifying Wind Turbine-Induced Seismic Noise Near Eskdalemuir: A Statistical Modelling Approach

Ridhima Gupta (S2757773) - Station G, 11:20; School of Mathematics, University of Edinburgh, Edinburgh, U.K.

Introduction

Motivation & Policy Setting: Eskdalemuir **(EKA; IMS/CTBT)** monitors **0.5–8 Hz**; **UK rules**: no turbines within **10 km**, and within **50 km** the *combined* motion must be **<0.336 nm RMS**; turbines can elevate energy in this band.

Objectives:

- Separate turbine-driven uplift via matched *background* vs *operational* windows.
- Quantify variation with *wind speed* and *wind direction* (orientation).
- Evaluate headroom relative to the **0.336 nm RMS** cap.
- Offer actionable guidance on monitoring and siting near **EKA**.

Study Site & Dataset Overview

Location: Eskdalemuir array, southern Scotland; all records come from borehole **WS12** only (no relocation or instrument swap).

Phases:

- **Background:** pre-build — turbines absent.
- **Operational:** post-build — turbines installed and producing power.

Data streams:

Time domain: vertical ground velocity $v(t)$ at **1s**

Frequency domain: PSD $S_x(f) \approx$ **0.01–10 Hz**; FDWF accentuates **0.5–8 Hz**.

Ancillary Fields: Wind speed (m/s), wind direction (deg/rad), turbine status (0/1); Date/Hour/Time.step. Co-registered and aggregated to 10-min windows for modelling.

Modelling Methodology (FDWF → Energy → GAM)

Pre-processing:

- *PSD construction:* from 1 s vertical velocity $v(t)$, compute $S_x(f)$ (m^2/Hz) on a shared grid ($\approx$ 0.01–10 Hz; fixed Δf).
- *FDWF reweighting:* emphasise the EKA band via $\tilde{S}_x(f) = w(f)\, S_x(f)$ (weights peak across **0.5–8 Hz**).
- *Scalar energy (per s):* $E_t = \sum_f \tilde{S}_x(f)\, \Delta f$ [m^2]; *Stabilised response:* $Y_t = \log_{10}(E_t + \varepsilon)$ with $\varepsilon = 10^{-12}$.
- *Time features:* `Hour` (0–23), `DateF` = Date + Hour/24.
- *Align & aggregate:* sync seismic/wind/status; model on **10-min** summaries.

> ***Why GAM?*** Captures **non-linear** wind effects, **cyclic** direction, and **day** random intercepts; remains interpretable.
> ***Model choice:*** compare nested GAMs (op, s(ws), s(ws) ⊠ op, s(dir)`"cc"`, s(DateF)`"re"`); select via AIC ↓ + diagnostics.

GAM as Final Model: Gaussian Y_t (identity); smooth, cyclic, and random-effect terms:

$$Y_t = \beta_0 + \beta_1\, \mathrm{Operational}_t + s_1(\mathrm{Wind.speed}_t) + s_2(\mathrm{Wind.speed}_t) \cdot \mathrm{Operational}_t + s_3^{\mathrm{cc}}(\mathrm{Wind.dir}_t) + b_{\mathrm{Date}(t)} + \varepsilon_t,$$

with $\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$ and $b_{\mathrm{Date}(t)}$ a day-level RE ($b_d \sim \mathcal{N}(0, \sigma_b^2)$).

- *Terms:* **Operational** shift; s_1 thin-plate smooth of wind speed; s_2 speed smooth × on/off; s_3^{cc} *cyclic* direction smooth; b_{Date} `bs="re"` (day RE).
- *Penalty & fit (REML):* maximise $\ell(\theta) - \sum_j \lambda_j \int [s_j''(x)]^2\, dx$; conservative k per covariate.
- *Implementation:* `mgcv::gam`; direction `bs="cc"`; day `bs="re"`.
- *Checks:* AIC, QQ, residual–fitted, ACF ⇒ assumptions adequate; mild serial dependence monitored.

Results & Conformity

Results: AIC-minimising GAM exposes strong directionality: peak energy under **south-westerlies (130–150°)**, with levels **higher when ON** across all bearings. **Wind speed** contributes little when **OFF**; under **ON** it yields a **modest, saturating** rise. Day-level random effects absorb small baseline shifts. On the energy scale ($\sim 10^{-24}\,\mathrm{m}^2$), **Operational 20% above Background**. Diagnostics (QQ/ACF) support good calibration and indicate weak serial dependence after 10-min aggregation. **0.5–8Hz operational uplift (SW 130–150°); RMS < 0.336 nm (max ≈ 0.177 *nm*) — compliant.**

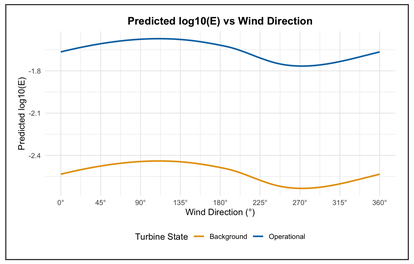

Figure 1 — Predicted $\log_{10}(E)$ vs wind direction (FDWF-weighted, 0.5–8 Hz)

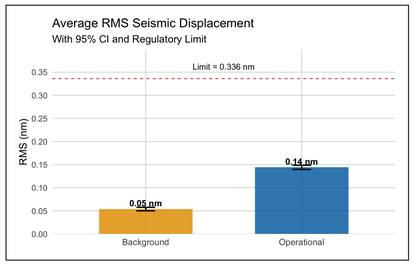

Figure 2 - Average RMS Seismic Displacement (0.5–8 Hz).

Conformity: RMS summary evidences uplift while remaining compliant: **Background** 0.012/0.018/0.024 nm (mean/95th/max); **Operational** 0.053/0.144/0.170 nm. All metrics lie below the **0.336 nm** limit; the peak 0.170 nm is **51%** of the threshold, leaving 0.166 nm headroom. ECDF/summary plots shift right when *ON*, yet the distribution stays within bounds—means and 95th percentiles remain sub-threshold; so no compliance concern under the observed conditions.

Limitations

- **WS12 single-site scope:** restricts generalisation to the wider array and nearby regional context.
- **Ops metadata sparse:** on/off only; no per-turbine load/position or layout no turbine-level attribution.
- **Residual dependence:** post–10-min aggregation autocorrelation; CIs may be optimistic — consider GAMM/AR terms or correlation.
- **Metric sensitivity:** energy and FDWF collapse the spectrum; findings hinge on the 0.5–8 Hz weighting assumptions applied. Sensitivity analyses and benchmarking are pending.

Future Work

To progress this study and guide sustainable turbine deployment, future work should explore:

- Scenario-based **cumulative impact modelling** to gauge **seismic implications** at **higher turbine density**.
- Adopt **spatiotemporal GAMs** or **Gaussian processes** to explicitly represent **spatial interference** and **terrain variation**.
- Employ **Bayesian GAMs** to propagate **uncertainty** in **uplift estimates**.
- **Turbine telemetry** integrated with the analysis for finer-grained attribution of **seismic signatures**.
- Undertake **multi-station validation** to evaluate **generalisability** and steer **regional infrastructure planning**.

Core Finding

Turbine operation elevates energy in **0.5–8 Hz**, peaking under **SW** winds; **20%** uplift. **RMS** remains below **0.336 nm** (≈0.177 nm); FDWF and GAM disentangle direction/speed effects for monitoring/planning.

Conclusion

Employing an FDWF-weighted energy measure and a GAM (interaction, cyclic direction, day RE), we identify turbine-driven uplift in the **0.5–8 Hz** window at EKA–WS12. The signal is small (energy $\sim 10^{-24}\,\mathrm{m}^2$); operational **RMS** reaches **0.177 nm** (~53% of the **0.336 nm** limit). No current compliance issue; continued, independent surveillance for cumulative impacts is strongly recommended routinely.

Full Report (PDF) + Code

Scan for the complete dissertation, figures, and Rmd

`https://publuu.com/flip-book/95/2091316`